RESEARCH OVERVIEW

CEED

Causal Expert–Evidence Distillation

Distilling a sparse MoE VLM teacher into a compute-matched dense student · 2026/08/19

Contents

Background

A sparse teacher, a dense student, and the gap between them.

The teacher–student pair

Teacher

gemma-4-26B-A4B-it

E0 E1 E2 E3
E4 E5 E6 E7

Every layer routes each token to a few of 128 experts — 26B total, 4B active.

Student

gemma-4-E4B-it

dense · router-free

Effective 4B via Per-Layer Embeddings — matched active compute, no routing at all.

Why this pair

Active-compute is matched, so any gain is attributable to **transferred structure**, not extra capacity.

The standard transfer signal: logit KD

Teacher forward pass answer tokens → **Top-k logits** per token → **KD loss** + CE on gold answer → **Student** backbone for every group

The gap

Logit KD teaches the student **what** the teacher said. It says nothing about **which internal computation** produced it — that structure is what CEED targets.

Motivation

The router predicts computation; it does not report it.

The router is an unreliable witness

What the router reports

Effective combine weight

predicted before compute runs

What actually happened

Causal Expert Attribution

measured — expert ablated, ΔlogP recorded

Empirical

Gating weight and ablated ΔNLL correlate only weakly (arXiv:2606.18304). Frequently-fired experts can be near-removable; some ablations **reduce** loss.

Consequence

Distilling router scores risks teaching the student the router's blind spots, not the teacher's division of labour.

Research question

Thesis

Does the **causally verified** division of computational labour in a MoE teacher — and the visual evidence driving it — transfer to a compute-matched dense student, when made an explicit training signal?

CEA

Per-token, per-expert **measured** causal attribution — not router probability.

ECC

How that attribution reorganises under targeted, controlled visual interventions.

All CEED machinery is removed after training — the deployed student is architecturally unchanged.

Related Work

Precedent map: what is taken, what supports the case, what remains open.

Precedent map

Taken

Sparse→dense KD (OneS); inactive-expert KD; counterfactual-image KD (VA-OPD); router-to-router KD; input-attribution KD (AD-KD, TSD); causal-abstraction distillation (**DIITO**).

Supports the case

Routing–attribution misalignment in pruning literature; dormant/correlational routing; per-expert ablation maps as interpretability; implicit-leakage risk (Shadow-MoE).

Remains open

Module-level causal attribution as a **distillation target**; coupling input interventions to internal attribution reorganisation; the compute-matched MoE→router-free setting.

Positioning against DIITO

DIITO (NAACL 2022) owns "distill causal structure, not just outputs" — CEED sits inside that family, not against it.

DIITO

causal abstraction

- Intervenes on **activations** (swaps hidden states)
- Requires layer/width-aligned dense pairs
- Matches counterfactual outputs

CEED

this work

- Intervenes on **inputs** (image regions) and **modules** (experts)
- No activation alignment needed
- Matches attribution structure, not activations

Why it matters

Activation interchange is ill-defined across the MoE⟶PLE architecture gap — CEED's input/module-level interventions are the reason this pair is tractable at all.

Our Methodology

CEA and ECC: two causally-grounded objects, distilled through three auxiliary losses.

Causal Expert Attribution (CEA)

Run teacher

record activated + near-miss experts

→

Ablate expert e

zero / mean-replace, re-run from layer m

→

ΔlogP

attribution of expert e

Near-miss experts

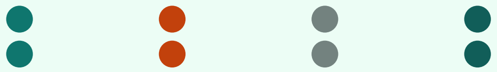
Top-k activated **plus** the next 2–3 by gating score — causally decisive experts can sit just below threshold.

Result

A per-(token, layer) **attribution vector** — comparable across contexts, unlike expert IDs.

Functional modes

raw attribution vectors



→

cluster / layer

8–32 modes

e.g. visual extraction, numeric manipulation, surface realization

Why modes

Coarser and more stable than raw vectors, invariant to expert-ID permutation — the supervision unit for the student's probe. Labels come from post-hoc analysis, never assumption.

Evidence–Computation Correspondence (ECC)

Relevant region

answer-critical, inpainted away

↓

Large attribution reorganisation

+ large output effect

Control region

matched size/texture, unrelated

↓

Near-zero reorganisation

+ near-zero output effect

The desired shape

Selectivity profile = large relevant effect, near-zero control effect. Coupling strength gates which tokens receive the coupling loss.

Four-cell diagnosis

	Output changed	Output stable
Attribution changed	**Strong coupling** selects tokens for coupling loss	**Internal compensation** genuine, not router noise
Attribution stable	**Ambiguous** rare — investigate	**Invariance** defines the control mask

Student objectives

Backbone: CE + top-k logit KD

every group shares this

↓

1. Mode probing

Removable linear probes predict per-token teacher mode. Deleted after training.

2. Selectivity matching

Match the student's per-token effect vector (relevant vs. control) to the teacher's.

3. Coupling consistency

On strong-coupling tokens, the probed mode shift must track the teacher's attribution shift.

Small gated weights (≈0.05–0.1 × CE), warm-up schedule (CE+KD first).

Architecture: the artifact store as the seam

Teacher
measured once, offline

→

Artifact store
CEA + ECC on disk, fingerprinted

→

Student training
teacher never loaded

→

Deployed E4B
probes removed, zero added cost

Extraction fingerprint

A content hash of layer set, ablation definition, combine-weight definition, thinking gate, and model revision — artefacts from different definitions can never silently mix.

Experiment Design

Twelve groups, one shared corpus and backbone, one varying signal.

The groups

Group	Auxiliary signal	Role
B0	none — zero-shot	floor
B1	none — SFT, no teacher	isolates teacher value
B2	CE + logit KD	**primary baseline**
B3	+ hidden-state projection KD	standard alternative
B4	+ VA-OPD reproduction	critical external baseline
B5	+ router-score distillation	routing vs. attribution ablation
C1 / C2	shuffled labels / mismatched layer	controls
E1	+ CEA mode probing	experimental component 1
E2	+ selectivity-profile matching	experimental component 2
E3 / E4	E1+E2 / +coupling consistency	additivity / full CEED

Phase 0 — go/no-go (teacher characterization)

- **0.1** Gating score vs. measured attribution correlation ⟶ expect $\rho \approx 0.3$–0.6 (misalignment)
- **0.2** Mode assignment MI vs. token-function, vs. router/hidden-state clusters ⟶ CEA must carry more
- **0.3** Mode stability across paraphrases and image pairs ⟶ modes track function, not token identity
- **0.4** ECC responsiveness: relevant vs. control reorganisation ⟶ expect ≥ 2–$3\times$

Fallback

$\rho > 0.9$ everywhere ⟶ CEA ≈ routing; abandon the pivot, fall back to a router-signal design (B5) with reduced claims.

Phase 1 — decision logic

- **E1 > B5 > B2** — gap concentrated on high-divergence tokens (+0.5–1.5 aggregate)
- **E2 > B4** on grounding selectivity + hallucination, even at comparable accuracy
- **E1 ≫ C1** — non-negotiable; shuffled-label parity would mean regularization, not transfer
- **E4 vs. E3** — coupling super-additivity, smallest and least certain effect

The paper survives on E1 / E2 + Phase 0

Even if E4 ≈ E3 (coupling is a null result), the mode-probing and selectivity-matching results stand on their own.

Phase 2 — fairness and robustness

Matched-FLOPs re-run

Grant B2 the ablation/intervention compute as extra vanilla-KD data. Edge should shrink ⅓ but survive.

Held-out generalization

Gains should persist, attenuated, outside the training task families.

Layer-mapping robustness

Repeat E1 under two alternative teacher↔student mappings.

Inference verification

Byte-identical E4B architecture; zero latency delta after probe removal.

Execution schedule and parallelism

Track	What	Weeks
Immediate	B0–B4, B5, shared intervention pipeline, corpus, teacher logits	1–2
Gated on pipeline	C1, C2, E1 (mode labels) · E2 (needs validated interventions)	4–5
Gated on conclusions	E3, E4 — only if Phase 0 passes go-criteria	6–12
Fairness	Phase 2: matched-FLOPs, generalization, robustness	12–16

Key point

Baseline checkpoints are frozen and reused everywhere — no baseline is ever retrained.

Implementation Status

Twenty-one tracked issues, from tooling to the full E4 pipeline.

Issue dependency map

Infra	01	02			
Corpus / B0–B2	03	04	05	06	
Baselines	07	08			
Phase 0	09	10	11	13	
Interventions	12	17			
E-track	14	15	18	19	20
Wrap-up	16	21			

done   ready-for-agent, not started

Status snapshot

Done

Workspace, CI, `run_group` spine, synthetic-teacher fixture, corpora (DocVQA/GQA/ChartQA), B0 zero-shot, artifact store + extraction, B1/B2 backbone.

In progress

07 (B1/B2 backbone) at 7/8 checklist items.

Not started

B3–B5; the full Phase 0 sweep (09–11, 13); intervention pipeline (12); the entire E-track (14–20); layer-mapping study and the Phase 1 table (16, 21).

Reading the map

Everything through B2 — the primary baseline — is built. The critical path from here is the intervention pipeline (12) and the Phase 0 sweep (09–11, 13): both gate the entire E-track.

Risks & Summary

What could go wrong, and what this work claims.

Risks and mitigations

Risk	Mitigation
Ablation cost	3 layers, top-k+2 experts, activation caching, answer tokens only
Attribution noise	Distill modes, not raw vectors; report Phase-0 reliability
Region-identification errors	Teacher-verified effect gaps; annotation-backed core results
PLE layer mapping	Empirical selection + two-mapping robustness report
Teacher errors	Correctness-flag gating
Auxiliary-loss interference	Warm-up schedule, small gated weights, early stopping

Contributions

1. Causal-attribution characterization of a production multimodal MoE
2. CEA as a distillation target — module-level **measured** attribution, not router scores or outputs
3. Evidence–computation coupling transfer — how structure reorganises under visual interventions
4. Inference-free transfer via removable probes — zero added cost
5. A grounding-selectivity evaluation protocol

Defensible statement

"We show that a multimodal MoE's router is an unreliable witness to its own computation and distill what the model measurably does instead — transferring evidence–computation correspondence into a router-free, compute-matched student at zero added inference cost."